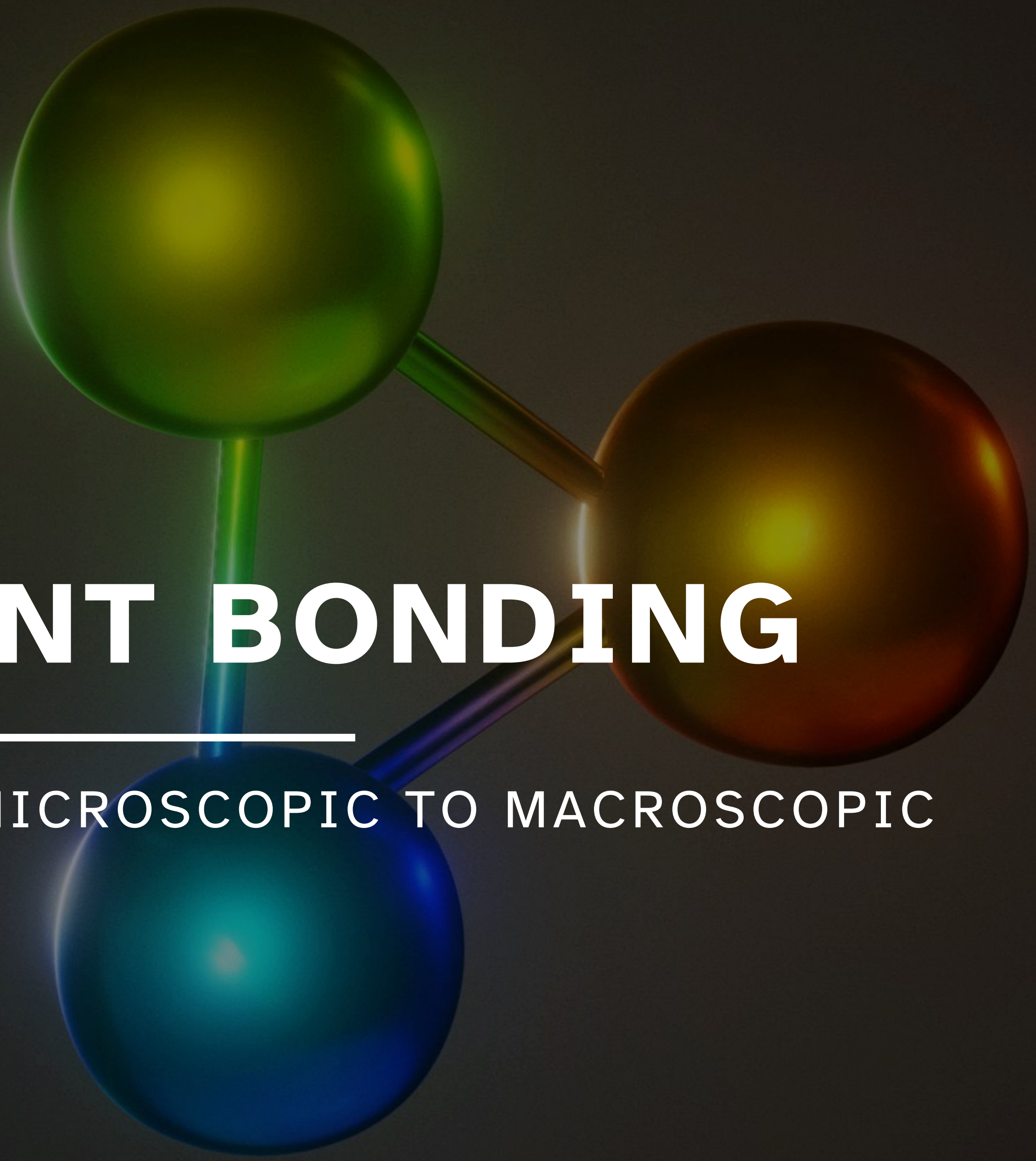


10A
COVALENT BONDING

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

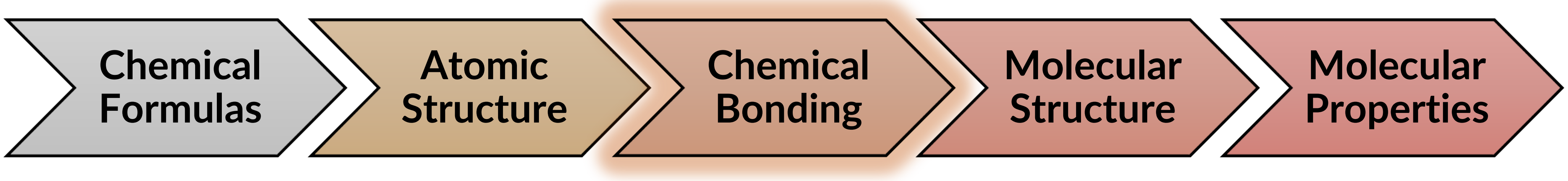
CHAPTER 10

1. Predict the number of bonds an atom is likely to form in a molecule.
2. Draw Lewis structures for molecules that follow the octet rule.
3. Define and apply electronegativity and its periodic trends.
4. Draw Lewis structures for molecules for which multiple valid Lewis structures exist (called resonance).
5. Draw Lewis structures for molecules that do not obey the octet rule by application of the typical exceptions exhibited by H, Be, B, and expanded octets.
6. Determine bond polarity using electronegativity values.
7. Describe and apply the relation between bond length and bond strength.
8. Apply bond enthalpies to calculate enthalpies of reaction.

Where did we start and where are we going?

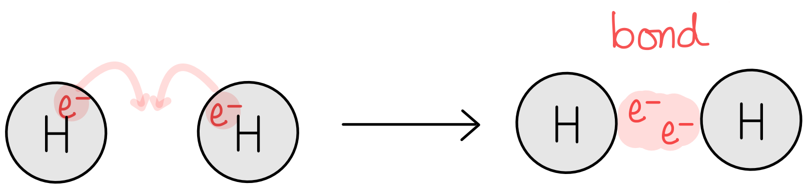
Chemistry in Context

What is our overall goal?



Formation of Covalent Bonds

10.1

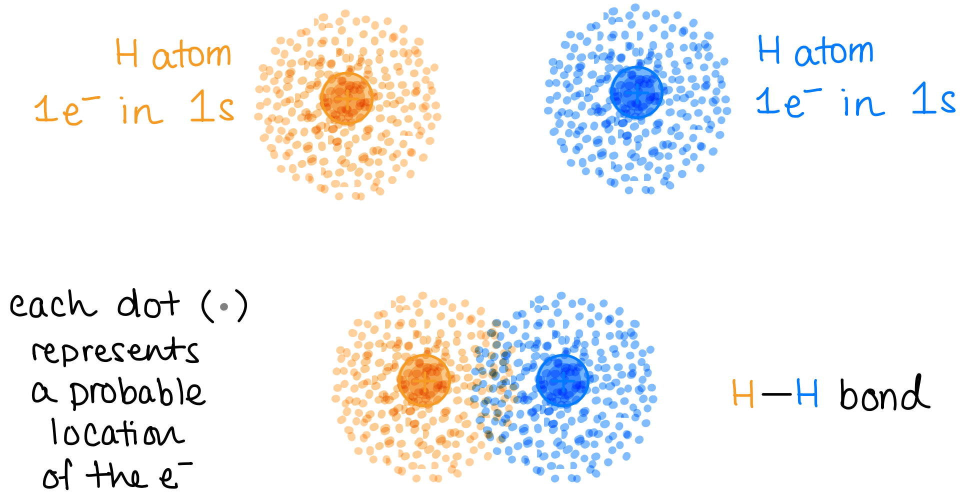


Covalent bonds

- Formed by the *sharing of electrons* in the space between two atoms.

How is it possible for these electrons to count for both atoms involved in the covalent bond?

- Electrons are small (quantum) particles.
- Electron properties have *wave-like* probabilities.
- Electron position can be thought of as “smeared” 3D charge clouds called orbitals.
- Electrons are *shared* when the orbitals (or “smeared” electron charge clouds) overlap.



Formation of Covalent Bonds

10.1

Octet (8 e^-) rule

- Atoms with **eight valence electrons are particularly stable** because they have a full valence shell with an electron configuration that is: s^2p^6
- In a molecule, atoms will form covalent bonds so that each atom attains a *complement* of 8 electrons ___THROUGH SHARING___.

especially true for $n=2$ row, but more a suggestion than a rule

Duet (2 e^-) rule

- For H and He, the valence shell is n = 1 and can only hold two electrons.
- They can share *only* two electrons in molecules to achieve a $1s^2$ configuration.

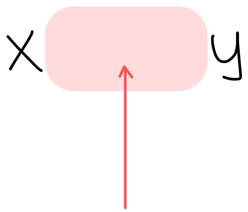
Resonance and Electronegativity

10.3 (we are not covering formal charges)

Electronegativity (EN)

- The ability of an atom to attract *shared* electrons to itself in a bond.
- Different numerical scales exist, but Pauling scale is most common.

- Trends in EN values depend on many factors: size, e^- config, IE, Z_{eff}, etc.



e^- exist in this space, but are they closer to X, Y, or evenly between?

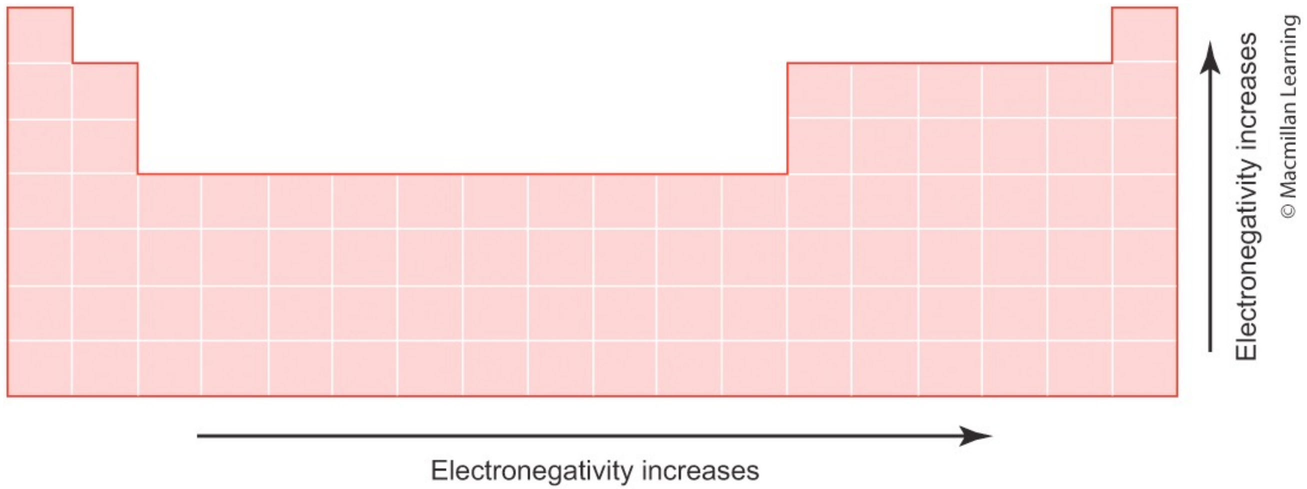


FIGURE 11.15 General electronegativity trend

Resonance and Electronegativity

10.3 (we are not covering formal charges)

An element with a *high electronegativity*:

- Pulls or attracts shared electrons closer to itself than an element with a lower electronegativity in a bond.

- less likely to share e⁻
- less likely to form many bonds

focus on trend!

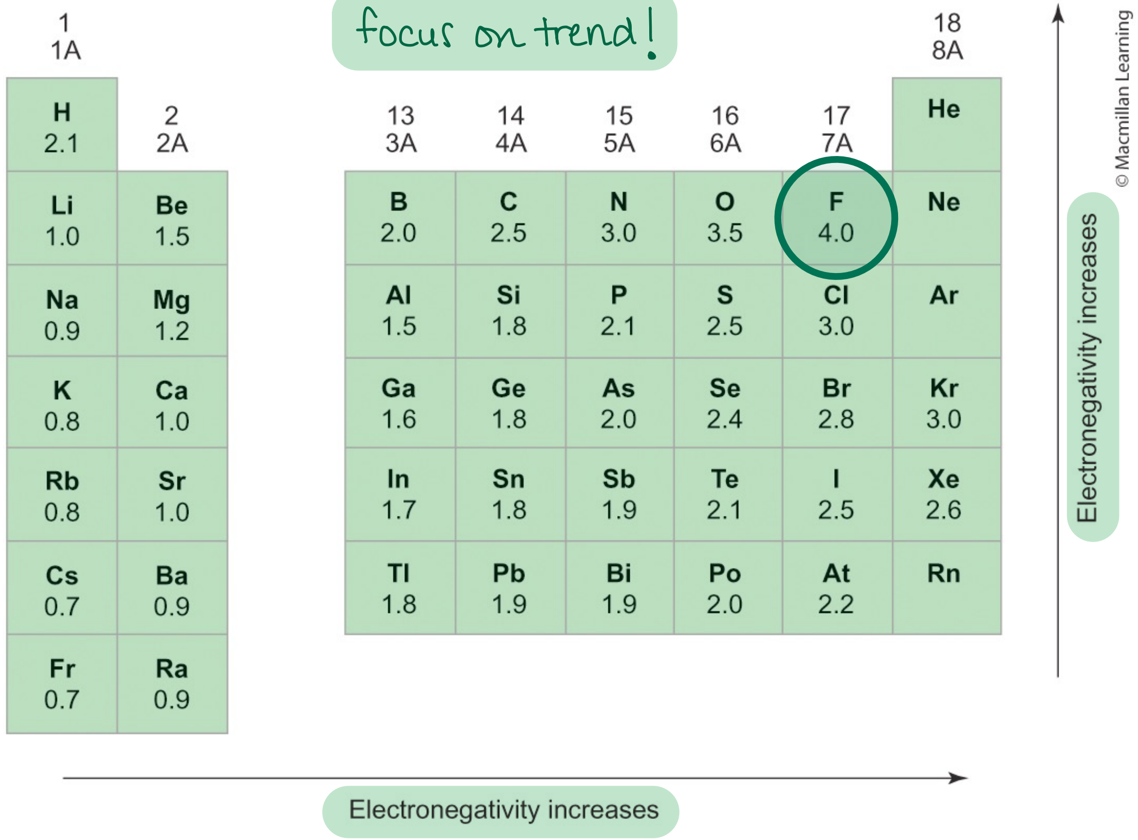


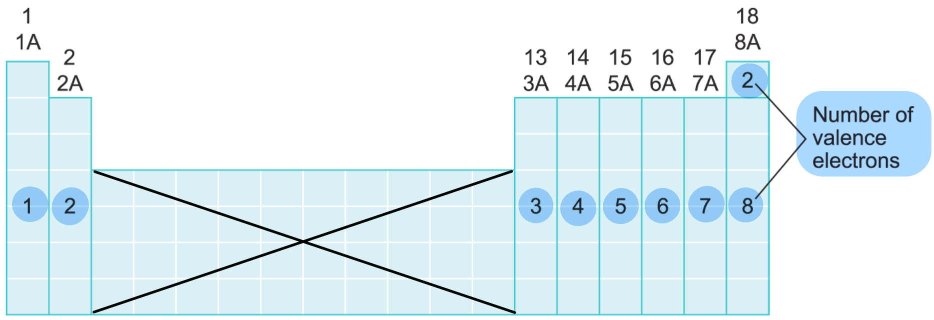
FIGURE 10.7 Pauling scale for electronegative values

Lewis Structures

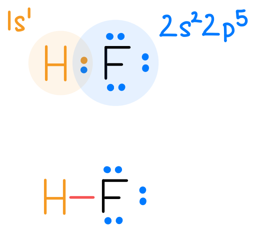
10.2

<u>Lewis structure</u>: *localized electron* model for the structure of covalent molecules

- *Only valence electrons*
- Total number of e^- in Lewis structure *must equal* total number of valence e^- from all atoms
- Satisfy octet (8 e^-) and duet (2 e^-) rules



Atom	# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Hydrogen **H**	1	2	0	2
Fluorine **F**	7	2	6	8



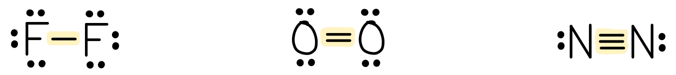
Formation of Covalent Bonds

10.1

Atoms can ***share*** different numbers of electrons to form different types of bonds.

Number of *shared* e^-	Bond Type	Representation	Example
2	Single bond	—	H_2, F_2
4	Double bond	=	O_2
6	Triple bond	≡	N_2

notes: area between atoms is shared;
each line is $2e^-$;
atoms end up with ns^2np^6 configuration

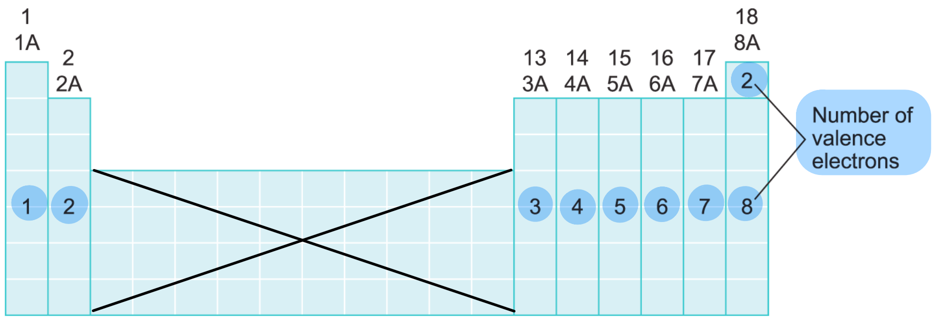


14 valence e^- 12 valence e^- 10 valence e^-

Lone pair: Set of 2 e^- that are ***not shared*** (not involved in bonding).

Lewis Structures

10.2

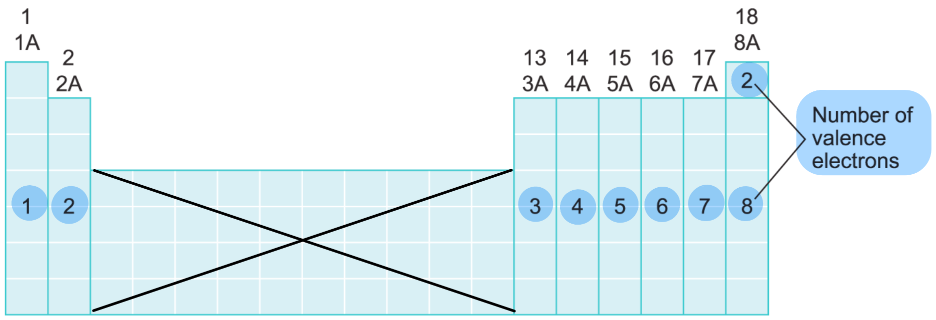


Method 1: Electron-Pushing

1. Count total number of valence e^- from all atoms.
2. Assign central atom and arrange all other atoms around it.
   - Central atom is *never* hydrogen.
   - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
3. Make skeleton: Use single bonds (–) to connect central atom to outer atoms.
4. Add lone pairs (··) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
5. Satisfy octet rule for central atom.
   a) If leftover *e*– (from step 1), add as lone pairs (··) to central atom.
   b) If *no* leftover *e*– (from step 1), re-assign lone pairs from outer atoms to convert single bonds (–) to double bonds (=) or triple bonds (≡).
6. Verify Lewis structure is valid: count total number of e^- in structure and check octet/duet rule.

Lewis Structures

10.2



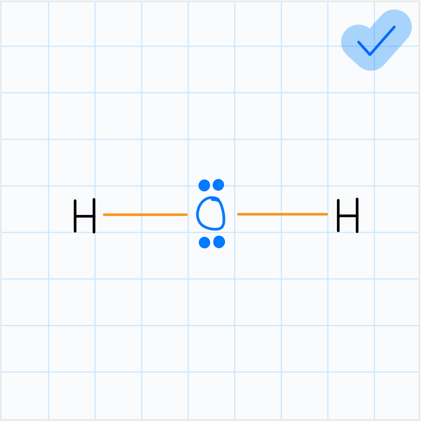
Method 2: Guess-and-Check

1. Count total number of valence e^- from all atoms.
2. Assign central atom and arrange all other atoms around it.
   - Central atom is *never* hydrogen.
   - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
3. Make skeleton: Use single bonds (—) to connect central atom to outer atoms.
4. Add lone pairs (··) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
5. Satisfy octet rule for central atom.
6. Count total number of e^- in structure drawn.
   a) If correct number of electrons as step 1, then valid Lewis structure.
   b) If *too many electrons* in structure, go back to step 3 and try double (=) and/or triple (≡) bonds.

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for H_2O.



$2(1e^-) + 6e^- = 8e^-$ total e^-

$- 4e^-$ 2 single bonds

$4e^-$

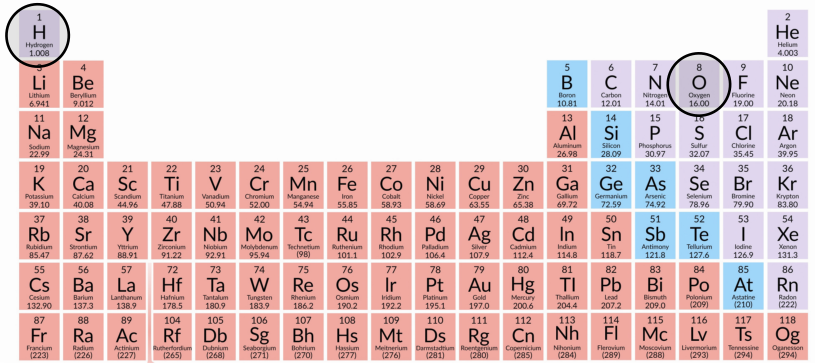
$- 0e^-$ octets on outer atoms

$4e^-$

$- 4e^-$ octet on central atom

$0e^-$

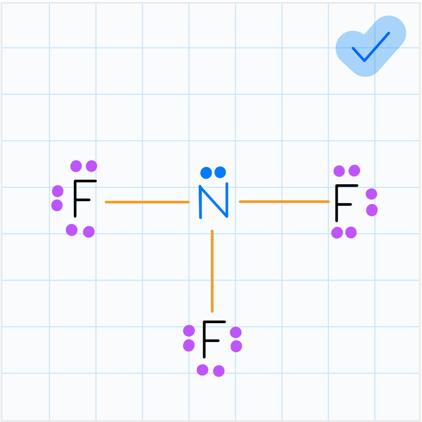
1. Count total # valence e^-
2. Pick central atom
3. Connect to central atoms using single bonds
4. Satisfy octet for outer atoms, or duet for H
5. Satisfy octet for central atom
   a) If leftover e^-, assign as lone pairs
   b) If no e^- left over, move lone pairs from outer atoms to double/triple bonds
6. Verify octet/duet rule and correct # total e^-



EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for NF_3.



$5e^- + 3(7e^-) = 26e^-$ total e^-

$-\ 6e^-$ 3 single bonds

$20e^-$

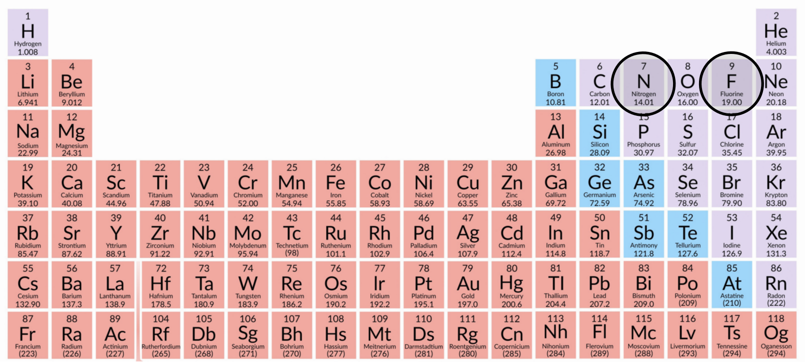
$-\ 18e^-$ octets on outer atoms

$2e^-$

$-\ 2e^-$ octet on central atom

$0e^-$

1. Count total # valence e^-
2. Pick central atom
3. Connect to central atoms using single bonds
4. Satisfy octet for outer atoms, or duet for H
5. Satisfy octet for central atom
   a) If leftover e^-, assign as lone pairs
   b) If no e^- left over, move lone pairs from outer atoms to double/triple bonds
6. Verify octet/duet rule and correct # total e^-



EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.4

Draw a valid Lewis structure for the carbonate anion (CO_3^{2-}).

2 extra e^-

$4 + 3(6) + 2 = 24\ e^-$ total e^-

$-\ 6\ e^-$ 3 single bonds

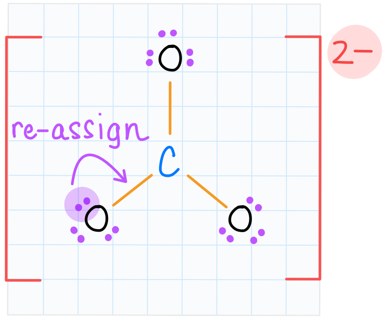
$18\ e^-$

$-\ 18\ e^-$ octets on outer atoms

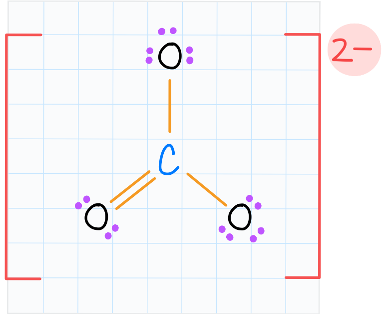
$0\ e^-$ out of e^- to fulfill octet on central atom!

↳ enact step 5b

✗

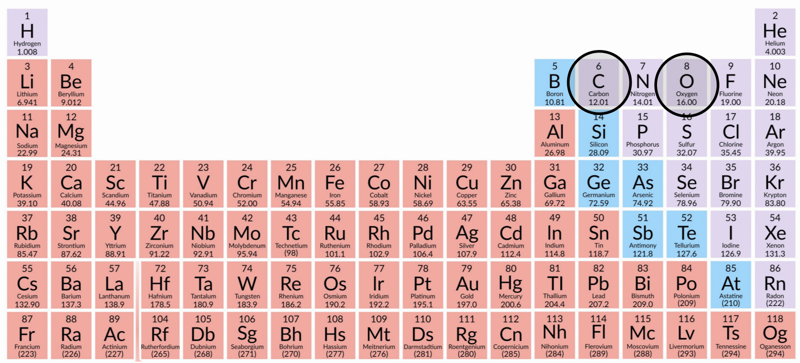


✓



for ions:
use square brackets and put overall charge or structure assumed to be neutral!

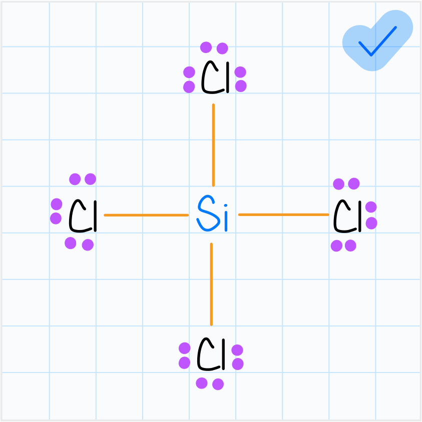
1. Count total # valence e^-
2. Pick central atom
3. Connect to central atoms using single bonds
4. Satisfy octet for outer atoms, or duet for H
5. Satisfy octet for central atom
    a) If leftover e^-, assign as lone pairs
    b) If no e^- left over, move lone pairs from outer atoms to double/triple bonds
6. Verify octet/duet rule and correct # total e^-



IN-CLASS QUESTION 1

Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for silicon tetrachloride ($SiCl_4$).



$4e^- + 4(7e^-) = 32e^-$ total e^-

$-\ 8e^-$ 4 single bonds

$24e^-$

$-\ 24e^-$ octets on outer atoms

$0e^-$

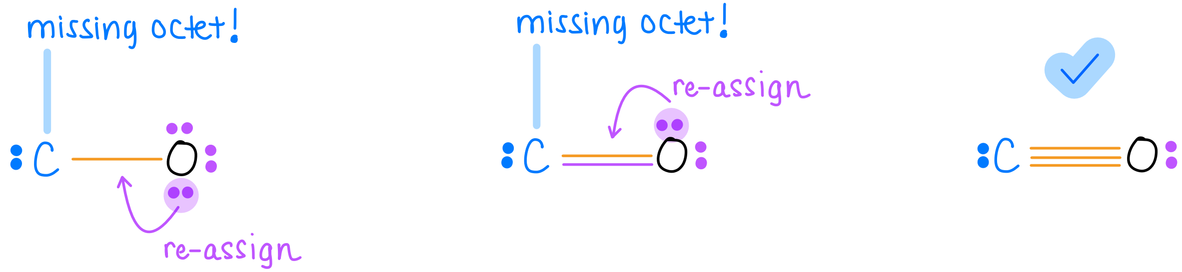
What is the total number of valence electrons in your Lewis structure? $32\ e^-$

What is the identity of the central atom? Si

IN-CLASS QUESTION 2

Learning Objective(s): 10.1, 10.2

Draw the Lewis structure for carbon monoxide (CO).



$4e^- + 6e^- = 10e^-$ total e^-

$-\ 2e^-$ 1 single bond

$8e^-$

$-\ 6e^-$ octets on outer atom

$2e^-$ out of e^- to fulfill octet on central atom!

↳ enact step 5b twice

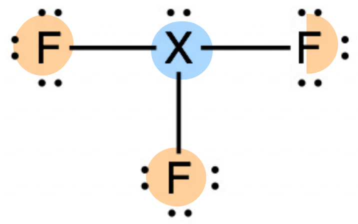
Complete the table below to describe the structure and bonding.

Atom	# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Carbon **C**	4	6	2	8
Oxygen **O**	6	6	2	8

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Which elements could be X in the Lewis structure shown?

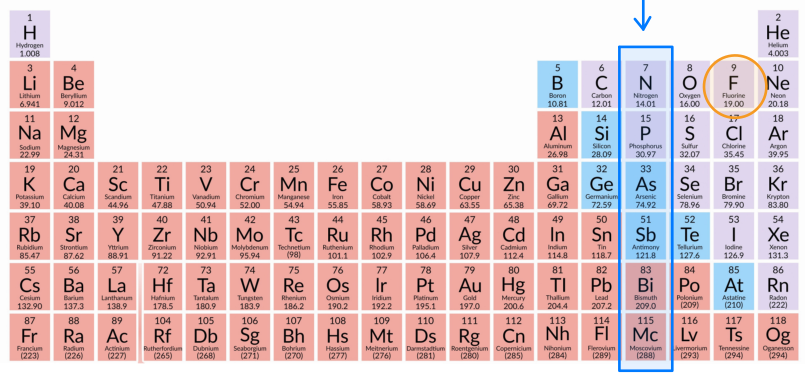


Select all that apply.

- [ ] C
- [x] N
- [ ] O
- [ ] F
- [ ] Si
- [x] P

(valence e^- on X) + 3(7 e^-) = 26 total e^-

valence e^- on X = 5 e^-

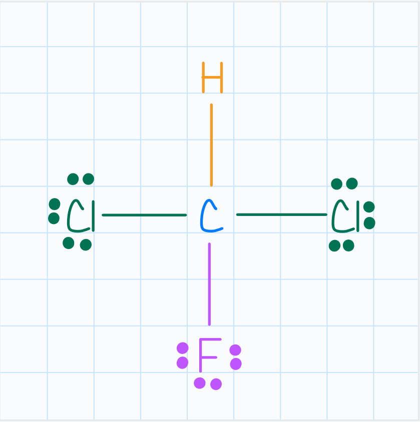


EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for $CHCl_2F$. $4e^- + 1e^- + 2(7e^-) + 7e^- = 26$ total e^-

Which statement is **true** regarding the atoms in this molecule?



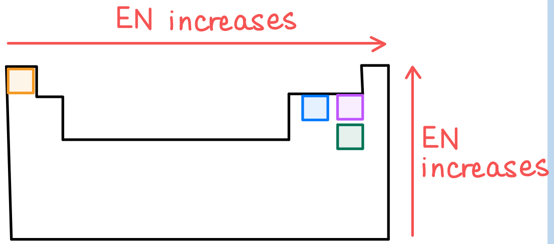
❑ In the bond between carbon and hydrogen, hydrogen attracts electrons ~~more~~ strongly than carbon.

❑ In the bond between carbon and chlorine, carbon gets a ~~larger~~ share of the electron pair than chlorine.

largest EN

☑ Fluorine has the greatest ability to attract electrons to itself in a bond.

❑ Carbon is the ~~most~~ electronegative atom.

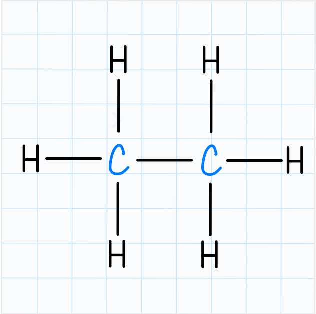


EXAMPLE PROBLEM

Learning Objective(s): 10.2

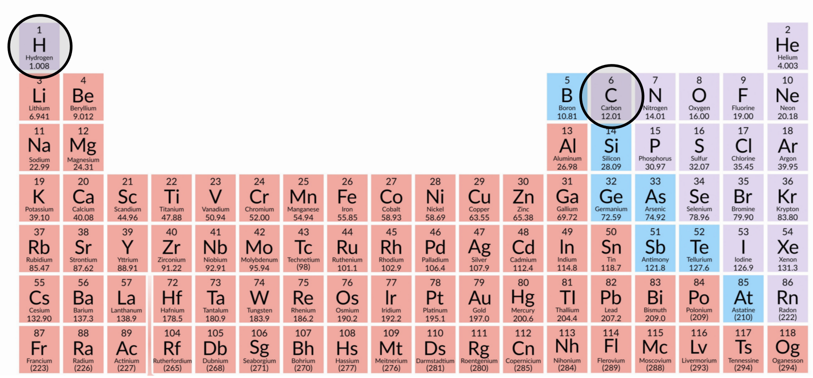
Draw the Lewis structure for C_2H_6.

$$2(4e^-) + 6(1e^-) = 14e^-$$



this is only possible structure!

we call both C atoms "central" atoms!

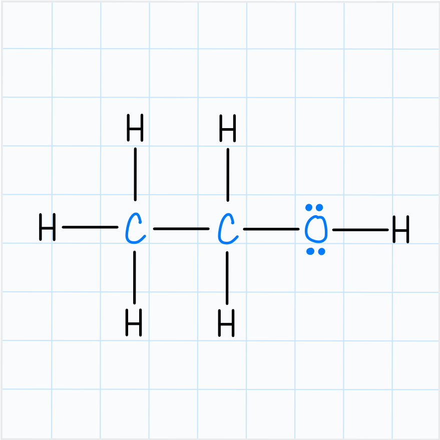


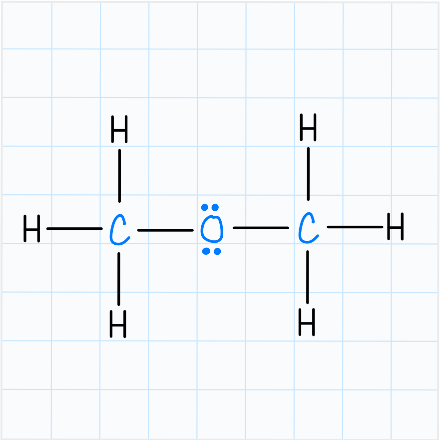
EXAMPLE PROBLEM

Learning Objective(s): 10.2

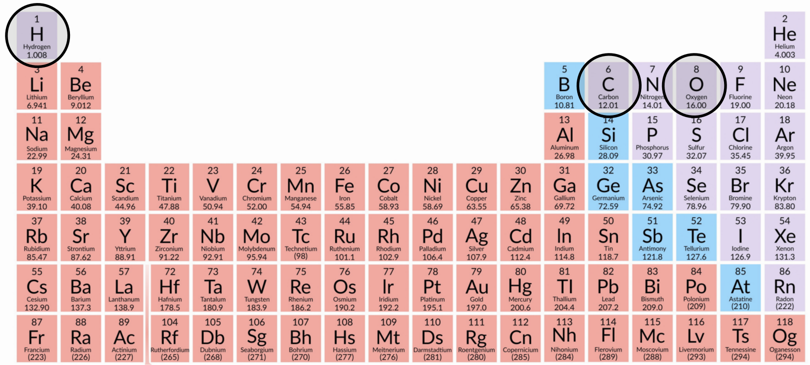
Draw two *different* Lewis structures for a molecule with the formula C_2H_6O.

$$2(4e^-) + 6(1e^-) + 6e^- = 20\ e^-$$





These are two different compounds with different properties!



BONUS QUESTION

Assessed on: Summer 2021, Exam 2

Draw the Lewis structures for CO_2, H_2O, and O_2.

a) Which molecule has the greatest number of electrons in shared bonds?

b) Which molecule has the smallest number of lone pair electrons?